

Kinematics

↳ The branch of mechanics which deals with the study of a body without cause of its motion is called kinematics.

• Rest and Motion

> Rest

↳ A body is said to be at rest if it does not change its position with respect to its surrounding.

> Motion

GSS_Notes

↳ A body is said to be in motion if it changes its position with respect to the / its surrounding.

Qn Can a body be regarded in the state of rest as well as in motion at the same time? Give an example.

Ans Rest and motion are the relative terms. So a body can be regarded in the state of rest as well as in motion at the same time. For example: a man sitting in a moving bus is at rest with respect to the driver but he is in motion with respect to the person standing outside the road.

- Distance and displacement

- Distance

↳ The length of actual path travelled by a body is called distance. It is a scalar quantity.

- Displacement

↳ The shortest distance between initial and final position of a body is called displacement. It is a vector quantity.

Qn What are the distance and displacement of a body moving in a semicircle of radius R ?

→ Distance travelled = πR
Displacement = $2R$



Qn Can a body have zero displacement but non zero distance travelled?

Ans Yes, a body can have zero displacement but not zero distance travelled.

When a body returns to the initial position after travelling some distance then displacement is zero but distance travelled is not zero.

- Speed and velocity

- Speed

↳ The rate of change of distance with respect to time is called speed.

$$\therefore \text{Speed} = \frac{\text{Distance travelled}}{\text{time}} = \frac{D}{T}$$

Its SI unit is m/s. It is a scalar quantity.

- Velocity

→ The rate of change of displacement of a body with respect to time is called velocity.

$$\therefore \text{Velocity} = \frac{\text{Displacement}}{\text{Time}} = \frac{s}{t}$$

Its SI unit is ms⁻¹. It is a vector quantity.

• Types of Velocity

i) Uniform Velocity

- A body is said to be in uniform velocity if it covers equal displacement in equal interval of time.

ii) Non-uniform or variable velocity

- A body is said to be in non-uniform velocity if it covers unequal displacement in equal interval of Time.

• Types of non-uniform velocity

Ⓐ Average Velocity

- It is defined as ratio of total displacement to the total time taken.

$$\text{i.e. Avg Velocity} = \frac{\text{Total displacement}}{\text{total time}}$$

Let S_1 and S_2 be the displacement of a body in time t_1 and t_2 respectively then

$$V_{av} = \frac{S_1 + S_2}{t_1 + t_2}$$

Note: (i) If time is equally divided then

$$V_{av} = \frac{V_1 + V_2}{2}$$

ii) If distance is equally divided then

$$V_{av} = \frac{2V_1 * V_2}{V_1 + V_2}$$

iii) For Uniform motion

$$V_{av} = V_{ins}$$

GSS Notes

(b) Instantaneous velocity

→ The velocity of a body at a particular time is called instantaneous velocity. It is given by limiting value of change in displacement per unit change in time.

$$\text{i.e.} - V_{ins} = \lim_{\Delta t \rightarrow 0} \frac{\Delta S}{\Delta t} = \frac{dS}{dt}$$

• Displacement time graph

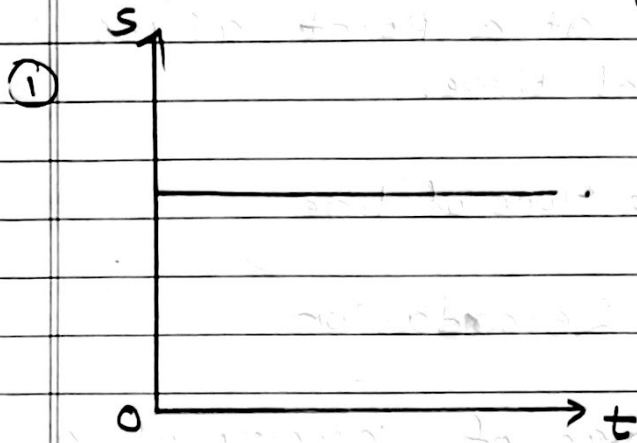


fig (a) : Zero velocity

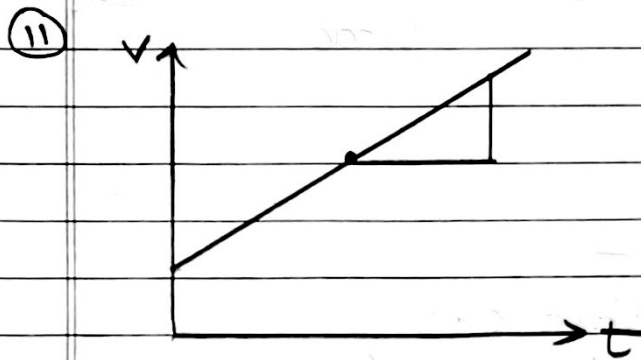


fig (b) : Uniform velocity

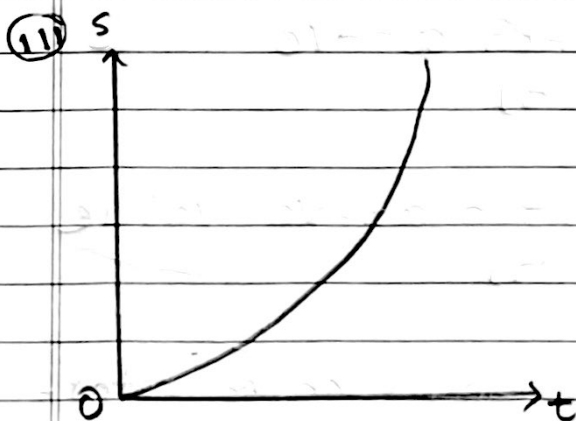


fig (c) : Non-Uniform velocity.

Note: In displacement - time graph, the slope of line at a point gives velocity at that time.

i.e - Velocity = Slope of line

- Acceleration and Retardation

→ The rate of change of velocity of a body with respect to time is called acceleration.

i.e - acceleration = $\frac{\text{change in velocity}}{\text{time}}$

$$\therefore a = \frac{V - u}{t}$$

Case - (I) : if $V = u$ then $a = 0$

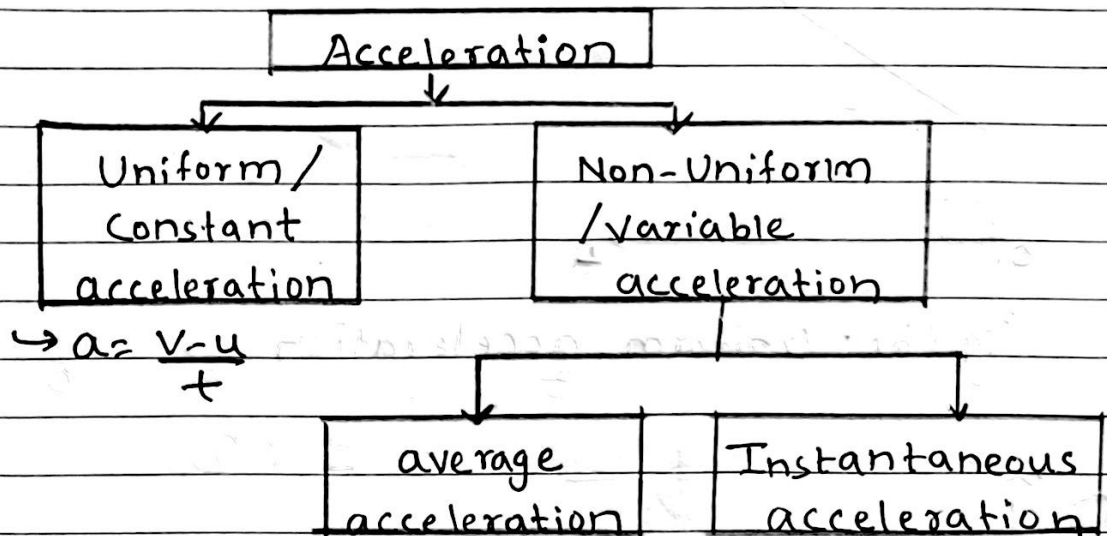
i.e - acceleration of a uniformly moving body is zero.

case - (II) : if $V > u$ then $a = +ve$ value
(i.e acceleration)

case - (III) : if $V < u$ then $a = -ve$ value
(i.e - retardation)

→ The negative acceleration is called retardation.

• Types of acceleration



$$\Rightarrow a_{ins} = \frac{dv}{dt}$$

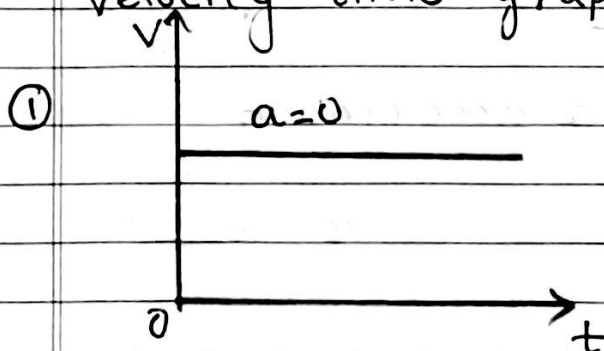
GSS_Notes

* Instantaneous acceleration

→ The acceleration of a body at a particular time is called instantaneous acceleration.

$$a_{ins} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

• Velocity-time graph



fig(a)- Zero acceleration

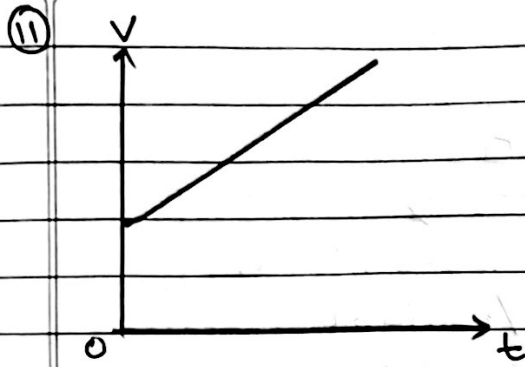


fig (b): Uniform acceleration

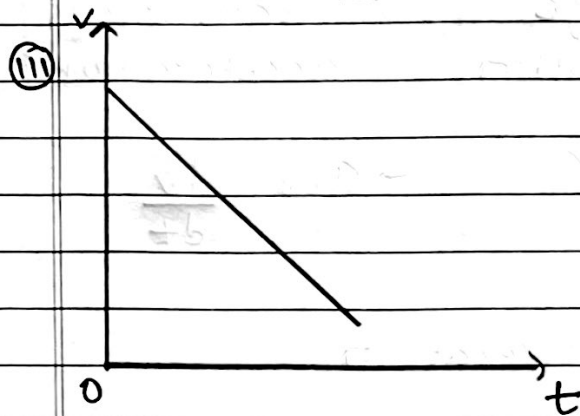


fig (c): Uniform retardation

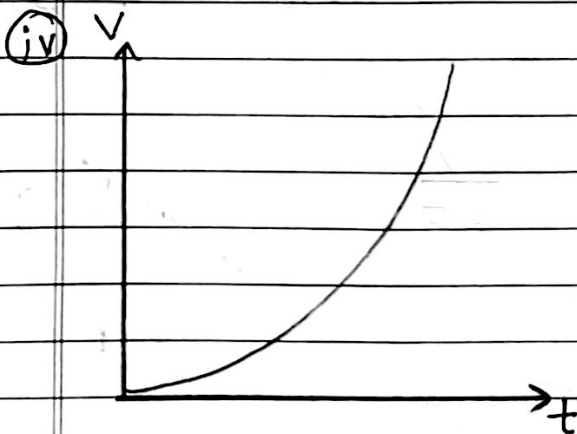


fig (d) - Non - Uniform acceleration

Note:

① In V-t graph, The slope of line at a point gives acceleration at that point.

i.e - acceleration = Slope of Line

② In V-t graph, Area under the line gives displacement.

i.e - Displacement = Area under the line

• Equations of motion for uniformly accelerated body

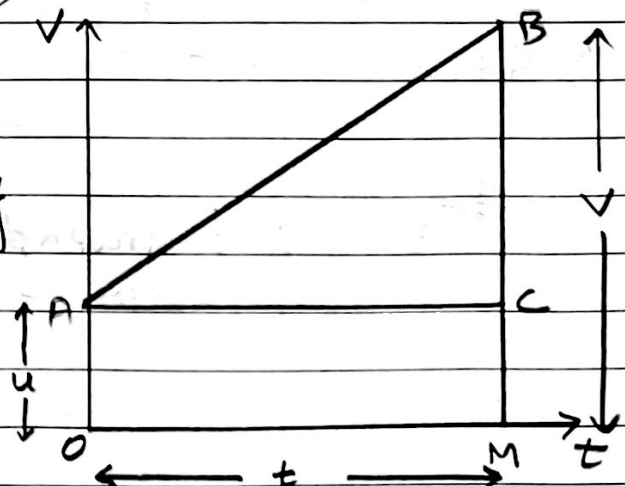
① $V = u + at$

② $S = ut + \frac{1}{2}at^2$

③ $V^2 = u^2 + 2as$

~~Deriv~~ Derivation From velocity-time graph

- The Velocity-time graph for a uniformly accelerated body is represented by the line AB.



In the figure:

Initial velocity (u) = OA

final velocity (v) = BM
time taken (t) = OM

① $V = ut + at$

Proof

In $V-t$ graph, We have

acceleration (a) = slope of line AB

$$\text{or, } a = \tan \theta$$

$$\text{or, } a = \frac{BC}{AC}$$

$$\text{or, } a = \frac{BM - CM}{OM}$$

$$\text{or, } a = \frac{BM - OA}{OM}$$

$$\text{or, } a = \frac{v - u}{t}$$

$$\text{or, } at = v - u$$

$$\text{or, } v = ut + at$$

$$\therefore v = ut + at$$

② $S = ut + \frac{1}{2} at^2$

Proof

In $V-t$ graph, we have, ~~displacement~~
displacement (s) = area under the
line AB

$$\text{or, } S = \text{Area of rectangle OACM} + \text{Area of triangle ABC}$$

$$\text{or, } S = OA \times OM + \frac{1}{2} AC \times BC$$

$$\text{or, } S = ut + \frac{1}{2} OM \times (BM - CM)$$

$$\text{or, } S = ut + \frac{1}{2} \times t \times (v - u)$$

$$\text{or, } S = ut + \frac{1}{2} \times t \times t \times \left(\frac{v - u}{t} \right)$$

$$\text{or, } S = ut + \frac{1}{2} \times t^2 \times a \quad \left[\because a = \frac{v - u}{t} \right]$$

$$\text{or, } S = ut + \frac{1}{2} at^2$$

$$\therefore S = ut + \frac{1}{2} at^2$$

GSS Notes

$$\textcircled{3} \quad v^2 = u^2 + 2as$$

Proof.

In v-t graph, we have
displacement (S) = area under the line
AB

$$\text{or, } S = \text{Area of trapezium OABM}$$

$$\text{or, } S = \frac{1}{2} \times \text{Sum of parallel sides} \times h$$

$$\text{or, } S = \frac{1}{2} \times (OA + BM) \times OM$$

$$\text{or, } S = \frac{1}{2} \times (u + v) \times t$$

$$\text{or, } 2S = ut + vt$$

$$\text{or, } 2as = at(u + v)$$

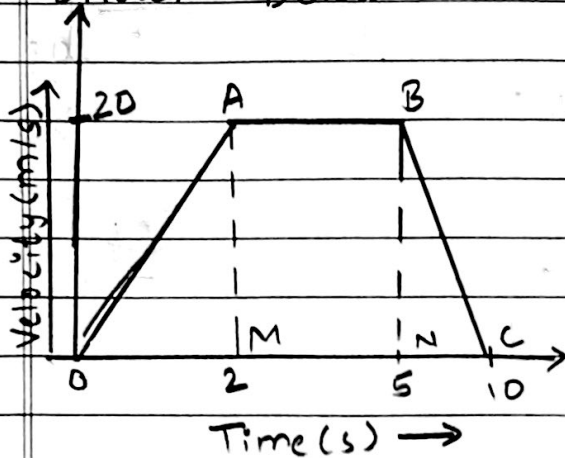
$$\text{or, } 2as = \frac{v - u}{t} \times t \times (u + v)$$

$$\text{or, } 2as = v^2 - u^2$$

$$\text{or, } v^2 = u^2 + 2as$$

$$\therefore v^2 = u^2 + 2as$$

Q.1) The velocity-time graph of an object moving along a straight line is as shown below:



Calculate:

- i) Acceleration of body between 0s ~~an~~ to 2s and 5s to 10s.

→ Soln,

Acceleration between 0s to 2s is given by $(a) = \text{slope of line OA}$

$$= \frac{AM}{OM}$$

$$= \frac{20}{2}$$

$$= 10 \text{ m/s}^2$$

Again,

Acceleration of body between 5s to 10s is given by $(a) = \text{Slope of line BC}$

$$= \frac{-13 \text{ N}}{5 \text{ s}}$$

$$= -\frac{20}{5}$$

$$= -4 \text{ m/s}^2$$

ii) Distance travelled by body between 0s to 5s and 0s to 10s.

→ Soln,

Distance travelled between 0s to 2s is given by $(s) = \text{Area under OAB}$

$$= \text{Area of } \triangle OAM + \text{Area of rectangle ABNM}$$

$$= \frac{1}{2} \times OM \times AM + MN \times AM$$

$$= \frac{1}{2} \times 2 \times 20 + 3 \times 20$$

$$= 20 + 60$$

$$= 80 \text{ m}$$

Again,

Distance travelled between 0s to 10s is given by $(s) = \text{Area under OABC}$

$$= \text{Area of trap. OABC}$$

$$= \frac{1}{2} \times (AB + OC) \times AM$$

$$= \frac{1}{2} \times (3+10) \times 20$$

$$= \frac{1}{2} \times 13 \times 20$$

$$= 130 \text{ m}$$

iii) What is the nature of motion of body between 2s to 5s?

→ Constant velocity i.e. - Zero acceleration.

② A car travelling with a speed of 15 m/s is braked and it slow down with uniform retardation. It covers a distance of 88m as its velocity reduces to 7 m/s. Calculate:

→ retardation of car.

→ Soln,

GSS_Notes

$$\text{Velocity (v)} = 7 \text{ m/s}$$

$$\text{distance (d)} = 88 \text{ m}$$

$$\text{time taken (t)} = ?$$

We know,

$$\text{Velocity} = \frac{\text{distance}}{\text{time taken}}$$

$$\text{or } 7 = \frac{88}{t}$$

$$\text{or } t = \frac{88}{7}$$

$$\text{or } t = 12.57 \text{ s}$$

i) retardation of car

→ Soln,

$$u = 15 \text{ m/s}$$

$$s = 88 \text{ m}$$

$$v = 7 \text{ m/s}$$

$$a = ?$$

Now,

$$v^2 = u^2 + 2as$$

$$\text{or, } 7^2 = 15^2 + 2 \times a \times 88$$

$$\text{or, } 49 = 225 + 176a$$

$$\text{or, } 176a = 49 - 225$$

$$\text{or, } 176a = -176$$

$$\text{or, } a = -1 \text{ m/s}^2$$

$$\therefore \text{retardation} = 1 \text{ m/s}^2$$

ii) further distance travelled by car before coming rest if it continues to slow down at same rate.

→ Soln,

$$u = 7 \text{ m/s}$$

$$v = 0$$

$$a = -1 \text{ m/s}^2$$

$$s = ?$$

We know that,

$$v^2 = u^2 + 2as$$

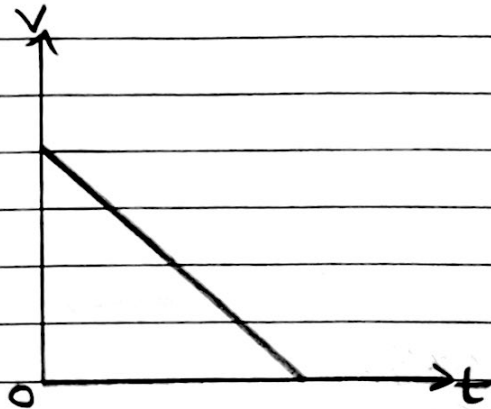
$$\text{or, } 0^2 = 7^2 + 2 \times -1 \times s$$

$$\text{or, } 0 = 49 - 2s$$

$$\text{or, } s = \frac{49}{2}$$

$$\text{or, } s = 24.5 \text{ m}$$

iii) Sketch the nature of velocity-time graph for the motion of car.



3. A ball is thrown vertically upward from ground with velocity of 20 m/s. calculate.

i) Maximum height reached by ball.

→ Soln,

$$u = 20 \text{ m/s}$$

$$v = 0 \text{ m/s}$$

$$g = -9.8 \text{ m/s}^2$$

$$h = ?$$

We know that,

$$v^2 = u^2 + 2as$$

$$\text{or, } 0^2 = 20^2 + 2gh$$

$$\text{or, } 0 = 400 + 2 \times -9.8 \times h$$

$$\text{or, } 0 = 400 - 19.6h$$

$$\text{or, } 19.6h = 400$$

$$\text{or, } h = \frac{400}{19.6}$$

$$h = 20.4$$

$$\therefore \text{height} = 20.4 \text{ m}$$

ii) Time taken by ball to return the ground.

→ let t be the time taken by the ball to reach to the maximum height.

then,

$$V = u + at$$

$$\text{or, } 0 = 20 + (-10)t$$

$$\text{or, } 10t = 20$$

$$\text{or, } t = 2\text{ s}$$

Then,

time taken by the ball to return to the ground is

$$T = \text{Ascending time} + \text{Descending time}$$

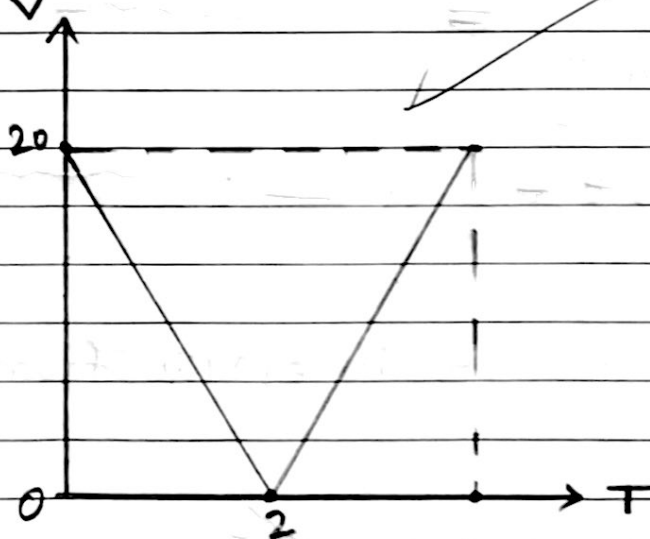
$$= t + t$$

$$= 2t$$

$$= 2 \times 2$$

$$= 4 \text{ sec.}$$

iii) Sketch the nature of velocity-time graph for the motion of ball.



4. An object is dropped from the top of the tower of height 156.2 m . and at the same time another object is thrown vertically upward with the velocity of 78.1 ms^{-1} from the foot of the tower, when and where the object meet?

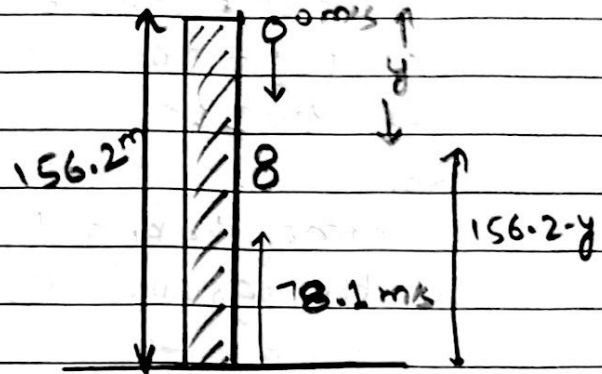
→ Soln,

For Dropped body,

$$h = y$$

$$u = 0\text{ ms}^{-1}$$

$$g = 9.8\text{ m/s}^2$$



$$S = ut + \frac{1}{2}gt^2$$

$$\text{or, } y = 0 \times t + \frac{1}{2}gt^2$$

GSS_Notes

$$\text{or, } y = \frac{1}{2}gt^2 \rightarrow (i)$$

For projected body,

$$S = ut + \frac{1}{2}at^2$$

$$156.2 - y = 78.1t + \frac{1}{2}gt^2 \rightarrow (ii)$$

Putting the value of y in eqn (ii) from eqn (i) We get.

$$156.2 - \cancel{\frac{1}{2}gt^2} = 78.1t - \cancel{\frac{1}{2}gt^2}$$

$$\text{or, } 156.2 = 78.1 t$$

$$\text{or, } t = \frac{156.2}{78.1}$$

$$\text{or, } t = 2 \text{ sec.}$$

$$\therefore t = 2 \text{ sec.}$$

Now,

Putting the value of t in eqn (i)

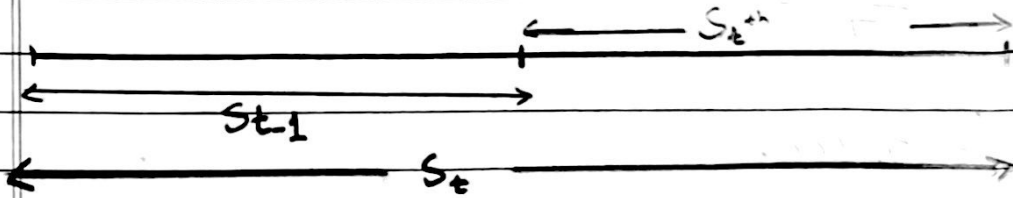
$$y = \frac{1}{2} \times 9.8 \times 2 \times 2$$

$$y = 19.6 \text{ m}$$

$$\therefore y = 19.6 \text{ m}$$

Hence the object meet below 20 m from the top of the tower after 2 s.

Distance travelled by a body in t^{th} second



Let S_t and S_{t-1} be the distance travelled by a body in t second and $(t-1)$ second respectively. Then distance travelled by the body in t^{th} second is given by.

$$\begin{aligned}
 S_{t^{\text{th}}} &= S_t - S_{t-1} \\
 &= ut + \frac{1}{2}at^2 - \left[u(t-1) + \frac{1}{2}a(t-1)^2 \right] \\
 &= ut + \frac{1}{2}at^2 - \left[ut - u + \frac{1}{2}a(t^2 - 2t + 1) \right] \\
 &= \cancel{ut} + \frac{1}{2}at^2 - \cancel{ut} + u - \frac{1}{2}at^2 + at - \frac{1}{2}a \\
 &= u + at - \frac{1}{2}a
 \end{aligned}$$

$$S_{t^{\text{th}}} = u + \frac{1}{2}a(2t-1)$$

✓.

S.Q. A body is dropped from a height 45 m.
Find the distance travelled by the body during the last second of its motion.

→ Soln,

$$h = 45 \text{ m} = S$$

$$u = 0 \text{ m/s}$$

$$a = 10 \text{ m/s}^2$$

$$S_{t^{\text{th}}} = ?$$

We know that,

$$S = ut + \frac{1}{2} at^2$$

$$\text{or, } 45 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$\text{or, } 45 = 5t^2$$

$$\text{or, } t^2 = 9$$

$$\therefore t = 3 \text{ sec}$$

Then,

$$S_{t^{\text{th}}} = u + \frac{1}{2} a (2t - 1)$$

$$= 0 + \frac{1}{2} \times 10 (2 \times 3 - 1)$$

$$= 5 \times 5$$

$$= 25 \text{ m}$$

Projectile

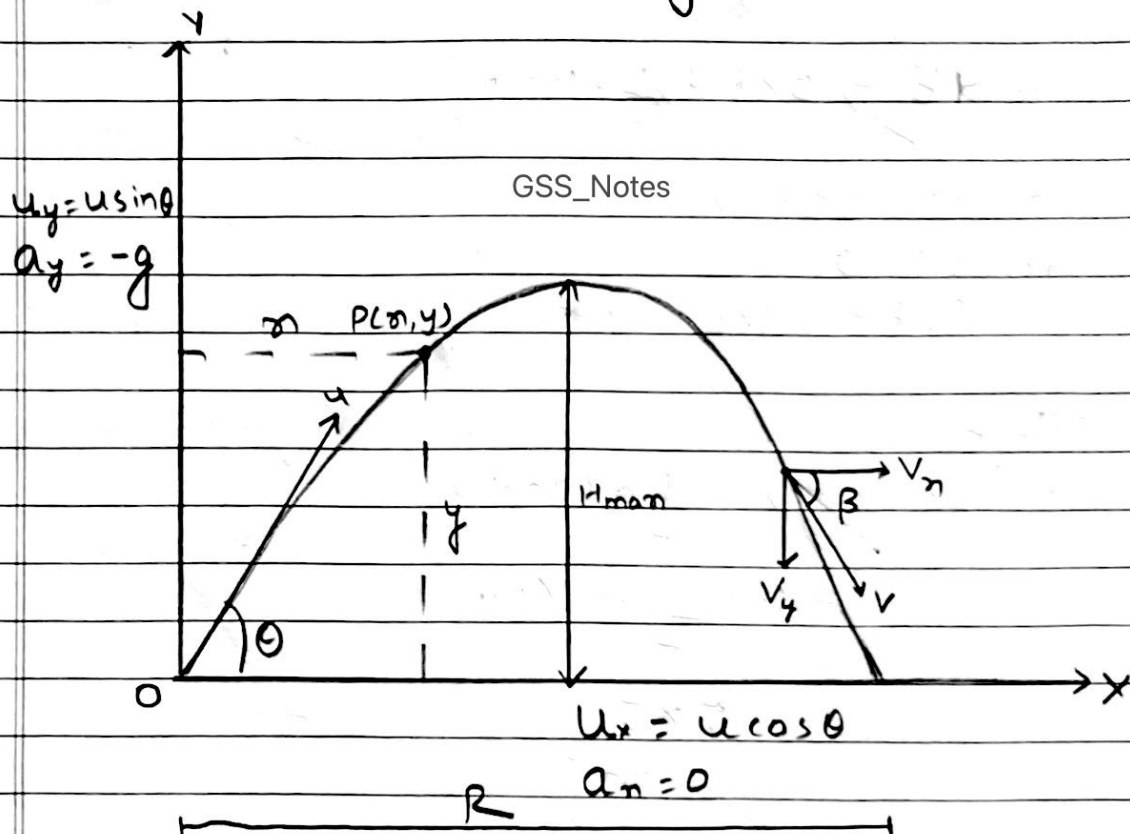
↳ If a body thrown into space or air moves only under the action of gravity then the body is called a projectile.

E.g - Bullet fired from a gun

- bomb dropped from aeroplane

- Stone thrown in air etc.

• Projectile fired from ground



Consider a body is fired from ground with velocity u making angle θ with the ~~low~~ ground as shown in figure.

The velocity u can be resolved into

two rectangular components $U_x = U \cos \theta$ and $U_y = U \sin \theta$ along x -axis and y -axis respectively.

The horizontal velocity of the projectile remains same at all the position. So horizontal acceleration (a_x) = 0. The Vertical velocity of the projectile changes continuously with time due to gravity. So, vertical acceleration (a_y) = $-g$.

① Trajectory of projectile

- The path followed by projectile in air is called ~~For~~ its trajectory.

Let x be the horizontal displacement and y be the vertical displacement of the projectile when it reaches from O to P in time t .

For horizontal motion,

$$S = ut + \frac{1}{2}at^2$$

$$\text{or, } x = U \cos \theta \times t + \frac{1}{2} \times 0 \times t^2$$

$$\text{or, } x = U \cos \theta \times t$$

$$\begin{aligned} \therefore S_x &= x \\ U_x &= U \cos \theta \\ V_x &= - \\ a_x &= 0 \\ t &= t \end{aligned}$$

$$\text{or, } t = \frac{x}{U \cos \theta} \rightarrow (i)$$

For vertical motion,

$$\therefore S_y = y$$

$$u_y = u \sin \theta$$

$$v_y = \sim$$

$$a_y = -g$$

$$t_y = t$$

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$\text{or, } y = u \sin \theta \times t + \frac{1}{2} \times -g \times t^2$$

$$\text{or, } y = u \sin \theta \times t - \frac{1}{2} g t^2 \rightarrow (ii)$$

Putting the value of t in eqn (ii) from eqn (i) We get,

$$y = u \sin \theta \times \frac{x}{u \cos \theta} - \frac{1}{2} g \times \left(\frac{x}{u \cos \theta} \right)^2$$

$$y = x (\tan \theta) - \frac{g}{2} \times \frac{x^2}{u^2 \cos^2 \theta}$$

$$\therefore y = x (\tan \theta) - \left(\frac{g}{2 u^2 \cos^2 \theta} \right) x^2 \rightarrow (iii)$$

This gives equation of trajectory of the projectile.

It is the equation of parabola. Hence the trajectory of the projectile is parabolic in nature.

(ii) Time of flight (T)

↳ The time spent by projectile in air is called its time of flight.

↳ It depends upon vertical motion of the

projectile.

Let 't' be the time taken by the projectile to reach to the maximum height.

Now,

$$\left[\begin{array}{l} V_y = u_y + a_y t \\ \text{or, } 0 = u \sin \theta - g t \\ \text{or, } g t = u \sin \theta \\ \text{or, } t = \frac{u \sin \theta}{g} \rightarrow (iv) \end{array} \right] \left[\begin{array}{l} \therefore S_y = x \\ a_y = u \sin \theta \\ V_y = 0 \\ a_y = -g \\ t = t \end{array} \right]$$

This gives ascending time.

Then, Time of flight = Ascending time + Decending time.

$$\text{or, } T = t + t$$

$$\text{or, } T = 2t$$

$$\text{or, } T = \frac{2 u \sin \theta}{g} \rightarrow (v)$$

(iii) Maximum height ($H_{\max}$)

→ It is the greatest vertical displacement of the projectile.

Now,

$$\left[\begin{array}{l} V_y^2 = u_y^2 + 2a_y S_y \\ \text{or, } 0^2 = u^2 \sin^2 \theta + 2(-g) H_{\max} \\ \text{or, } 0 = u^2 \sin^2 \theta - 2g H_{\max} \\ \text{or, } 2g H_{\max} = u^2 \sin^2 \theta \end{array} \right] \left[\begin{array}{l} S_y = H_{\max} \\ u_y = u \sin \theta \\ V_y = 0 \\ a_y = -g \\ t = - \end{array} \right]$$

$$\therefore H_{\max} = \frac{u^2 \sin^2 \theta}{2g} \rightarrow (6)$$

(iv) Horizontal Range (R)

→ The horizontal displacement of the projectile during its time of flight is called horizontal range.

Now,

$$S_x = ut + \frac{1}{2} at^2$$

$$\text{or, } R = u \cos \theta \cdot T + \frac{1}{2} 0 \times T^2$$

$$\text{or, } R = u \cos \theta \cdot T$$

$$\text{or, } R = u \cos \theta \cdot \frac{2u \sin \theta}{g}$$

$$\therefore R = \frac{u^2 \sin 2\theta}{g} \rightarrow \textcircled{7} [\because \sin 2\theta = 2 \sin \theta \cos \theta]$$

• Condition for maximum horizontal Range

GSS_Notes

From eqn (7)

$$R \propto \sin 2\theta \text{ [For constant } u]$$

For the maximum value of R, the value of $\sin 2\theta$ should be maximum.

$$\text{i.e. } \sin 2\theta = 1$$

$$\text{or, } \sin 2\theta = \sin^{-1}(1)$$

$$\text{or, } 2\theta = 90^\circ$$

$$\text{or, } \theta = \frac{90^\circ}{2}$$

$$\therefore \theta = 45^\circ$$

Hence the horizontal range becomes maximum when the angle of projection is 45° and the maximum horizontal range

is given by $R_{\max} = \frac{u^2}{g} \rightarrow (8)$

> Two angles of projection for the same horizontal range

- For the angle of projection θ , the horizontal range is given by

$$R_1 = \frac{u^2 \sin 2\theta}{g} \rightarrow (9)$$

Again For the angle of projection $(90-\theta)$, the horizontal range is given by

$$\begin{aligned} R_2 &= \frac{u^2 \sin 2(90-\theta)}{g} \\ &= \frac{u^2 \sin (180-2\theta)}{g} \\ &= \frac{u^2 \sin 2\theta}{g} \rightarrow (10) \end{aligned}$$

Here, $R_1 = R_2$. This shows that two angles of projection for the same horizontal range are θ and $(90-\theta)$.

⑤ Velocity of projectile at any instant

Let v be the velocity of projectile at any instant t .

The horizontal component of this velocity is given by

$$V_x = u_x + a_x t$$

$$\text{or, } V_x = u \cos \theta + 0 \times t$$

$$\therefore V_x = u \cos \theta \rightarrow (11)$$

and vertical component of this velocity is given by

$$V_y = u_y + a_y t$$

$$\text{or, } V_y = u \sin \theta + (-g)t$$

$$\therefore V_y = u \sin \theta - gt \rightarrow (12)$$

$$\left[\begin{array}{l} \therefore S_x = x \\ u_x = u \cos \theta \\ V_x = V_x \\ a_x = 0 \\ t = t \end{array} \right]$$

$$\left[\begin{array}{l} \therefore S_y = y \\ u_y = u \sin \theta \\ V_y = V_y \\ a_y = -g \\ t = t \end{array} \right]$$

Then,

Magnitude of velocity is given by

$$V = \sqrt{(V_x)^2 + (V_y)^2}$$

$$\therefore V = \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2} \rightarrow (13)$$

And the direction of velocity (V) is given by

$$\beta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$$

$$\therefore \beta = \tan^{-1} \left(\frac{u \sin \theta - gt}{u \cos \theta} \right) \rightarrow (14)$$

Q. A projectile is fired from ground with velocity 500 m/s at 30° to the horizontal. Find &

- (i) position of the projectile after two second.
- (ii) Time taken by the projectile to reach to the maximum height.

- (iii) Time of flight
- (iv) Maximum height
- (v) Horizontal range
- (vi) Minimum velocity of projectile to achieve the same horizontal range.
- (vii) Velocity of projectile after 3 seconds.

→ Soln,

$$u = 500 \text{ m/s}$$

$$\theta = 30^\circ$$

① $x = ?$, $y = ?$

For horizontal motion,

$$S_x = u_x t + \frac{1}{2} a_x t^2$$

$$\text{or, } x = u \cos \theta \times t + \frac{1}{2} \times 0 \times t^2$$

$$\text{or, } x = u \cos \theta \times t$$

$$\text{or, } x = 500 \times \cos 30^\circ \times 2$$

$$x = 500 \times \frac{\sqrt{3}}{2} \times 2$$

$$x = 250\sqrt{3} \times 2$$

$$x = 500\sqrt{3}$$

$$\therefore x = 866.025 \text{ m}$$

And,

For vertical motion,

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$\text{or, } y = u \sin \theta \times t + \frac{1}{2} (-g) t^2$$

$$\text{or, } y = u \sin \theta \times t - \frac{gt^2}{2}$$

$$\text{or, } y = 500 \times \sin 30 \times 2 - \frac{1}{2} \times 10 \times 2^2$$

$$\text{or, } y = 500 \times \frac{1}{2} \times 2 - \frac{1}{2} \times 10 \times 2 \times 2$$

$$\text{or, } y = 500 - 20$$

$$\therefore y = 480 \text{ m}$$

Hence, the position of projectile is 866.025 m away and 480 m above the point of projection.

(ii) Maximum height = ?

We have formula,

$$H_{\text{max}} = \frac{u^2 \sin^2 \theta}{2g}$$

$$= \frac{500^2 \times \sin^2 30}{2 \times 10}$$

$$= \frac{250000 \times 1}{2 \times 4}$$

$$= \frac{25000}{8}$$

$$= 3125 \text{ m}$$

(iii) Time of flight = ?

We know that,

$$\text{Time of flight (T)} = \frac{2u \sin \theta}{g}$$

$$= \frac{2 \times 500 \times \sin 30}{10}$$

$$= 100 \times \frac{1}{2}$$

$$= 50 \text{ sec}$$

(ii) Time taken by the projectile to reach the maximum height $h(t) = ?$

We have formula,

$$T = 2t \quad \text{Where,}$$

T is time of flight,

$$\Rightarrow \frac{50}{2} = t$$

$$\therefore t = 25 \text{ s}$$

(v) Horizontal range (R) = ?

By formula,

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$= \frac{500^2 \times \sin 2 \times 30}{10}$$

$$= \frac{250000 \times \sin 60}{1}$$

$$= 250000 \times \frac{\sqrt{3}}{2}$$

$$= 21650.63 \text{ m}$$

GSS_Notes

(vii)

→ Soln,

Time taken = 3 sec

So,

$$V = \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2}$$

$$\text{or, } V = \sqrt{(500 \cdot \cos 30)^2 + (500 \times \sin 30 - 10 \times 3)^2}$$

$$\text{or, } V = \sqrt{\left(500 \times \frac{\sqrt{3}}{2}\right)^2 + \left(500 \times \frac{1}{2} - 30\right)^2}$$

$$= \sqrt{(250\sqrt{3})^2 + (250-30)^2}$$

$$= \sqrt{(250\sqrt{3})^2 + 220^2}$$

$$= 485.69 \text{ m/s}$$

vi)

→ Soln,

$$U_{\min} = ?$$

$$R = 21650.63 \text{ m}$$

Since,

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$\text{or, } \frac{21650.63 \times 10}{\sin 2\theta} = u^2$$

$$\text{or, } \frac{216506.3}{\sin 2\theta} = u^2$$

Here,

$$u^2 \propto \frac{1}{\sin 2\theta} \text{ . so for the minimum}$$

value of u , the value of $\sin 2\theta$ should be maximum. i.e. $\sin 2\theta = 1$

So,

$$U_{\min}^2 = 216506.3$$

$$= \sqrt{216506.3}$$

$$\therefore U_{\min} = 465.30 \text{ m/s}$$

8. A baseball is thrown towards a player with an initial speed of 20 m/s and 45° with horizontal. At the moment ball is thrown player is 50 m from thrower. At what speed and in what direction must he run to catch the ball at same height at which it was released?

→ Soln,

$$u = 20 \text{ m/s}$$

$$\theta = 45^\circ$$

Now,

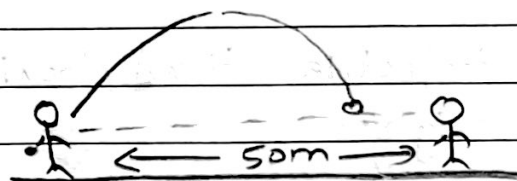
$$R = \frac{u^2 \sin 2\theta}{g}$$

$$= \frac{20^2 \times \sin 2 \times 45}{10}$$

$$= \frac{400 \times \sin 90}{10}$$

$$= 40 \times 1$$

$$= 40 \text{ m}$$



Here,

$R < 50 \text{ m}$ so the player should run ~~towards~~ towards the thrower.

Then,

$$\text{Time of flight} = \frac{2u \sin \theta}{g}$$

$$= \frac{2 \times 20 \times \sin 45}{10}$$

$$= \frac{40 \times 1}{10 \times \sqrt{2}}$$

$$= 2\sqrt{2}$$

$$= 2.82 \text{ s}$$

Now,

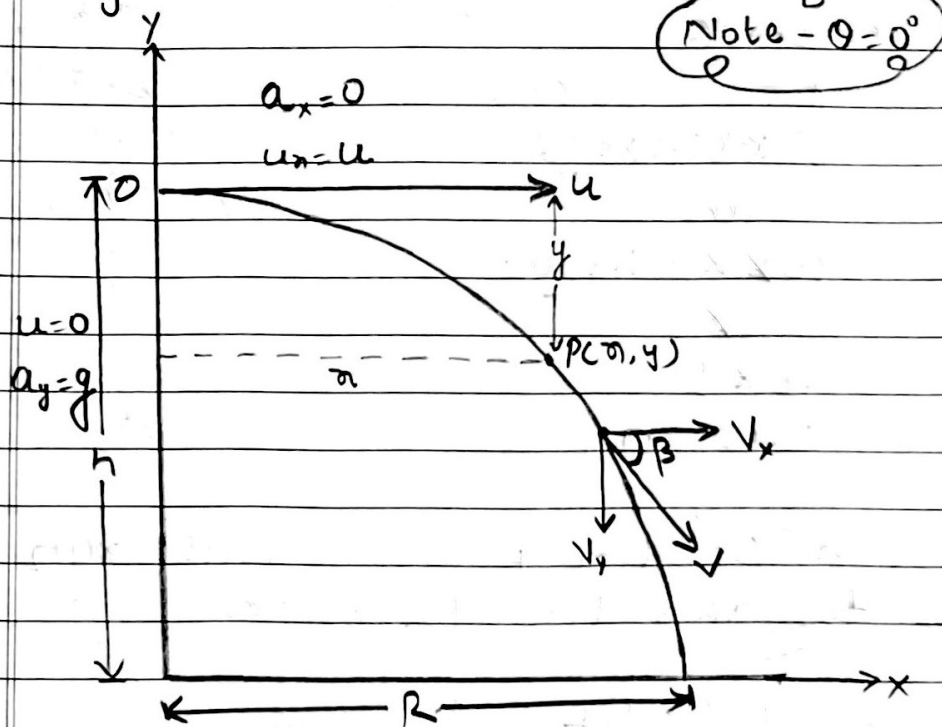
$$\text{Speed } (v) = \frac{s}{t}$$

$$= \frac{50-40}{2.82}$$

$$= \frac{10}{2.82}$$

$$= 3.54 \text{ m/s}$$

Projectile Fired horizontally from a height



Consider a projectile is fired horizontally from a height ' h ' with velocity ' u '.

The velocity ' u ' can be resolved into two rectangular components $u_x = u$ and $u_y = 0$ along x -axis and y -axis respec-

tively.

The horizontal velocity of the projectile remains same at all the positions.

So horizontal acceleration ($a_x = 0$).

The vertical velocity of the projectile changes continuously with time due to gravity. So vertical acceleration ($a_y = g$).

① Trajectory of projectile

- The path followed by projectile in air is called its trajectory.

Let x be the horizontal displacement and y be the vertical displacement of the projectile when it reaches through the position P in time 't'.

Now,

$$S_x = u_x t + \frac{1}{2} a_x t^2 \quad \text{GSS_Notes}$$

$$\text{or, } x = u_x t + \frac{1}{2} \times 0 \times t^2$$

$$\text{or, } x = u_x t$$

$$\therefore t = \frac{x}{u} \rightarrow (1)$$

And

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$\text{or, } y = 0 + \frac{1}{2} g t^2$$

$$\therefore y = \frac{1}{2} g t^2 \rightarrow (2)$$

Putting the value of t in eqn (2) from eqn (1) we get.

$$y = \frac{1}{2} g \left(\frac{x}{u} \right)^2$$

$$\therefore y = \left(\frac{g}{2u^2} \right) x^2 \rightarrow (3)$$

This ~~is~~ gives equation of trajectory of the projectile. It is the equation of parabola. So trajectory of the projectile is parabolic in nature.

(ii) Time of flight (T)

- Time Spent by projectile in air is called its time of flight.

Now,

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$\text{or, } h = 0 \times t + \frac{1}{2} g T^2$$

$$\text{or, } h = \frac{1}{2} g T^2$$

$$\text{or, } \frac{2h}{g} = T^2$$

$$\therefore T = \sqrt{\frac{2h}{g}} \rightarrow (4)$$

∴

Note: Time of flight only depends upon the height but it doesnot depend upon the initial velocity of the

projectile. So if two bodies are projected horizontally with different velocities from the same height then they reach to the ground at the same time.

(iii) Horizontal range (R)

- The horizontal displacement of projectile during its time of flight is called horizontal range.

Now,

$$S_x = u_x t + \frac{1}{2} a_x t^2$$

$$\text{or, } R = u \times T + \frac{1}{2} \times 0 \times T^2$$

$$\text{or, } R = uT$$

$$\therefore R = u \sqrt{\frac{2h}{g}} \rightarrow (5)$$

(iv) Velocity of projectile at any instant

- let 'v' be the velocity of the projectile at any instant 't'. Now, horizontal component of v is given by

$$V_x = u_x + a_x t$$

$$\text{or, } = u + 0 \times t$$

$$= u \rightarrow (6)$$

And, vertical component of v, is given by

$$\begin{aligned} V_y &= U_y + a_y t \\ &= 0 + gt \\ &= gt \rightarrow (7) \end{aligned}$$

Then,

$$\begin{aligned} V &= \sqrt{V_x^2 + V_y^2} \\ \therefore V &= \sqrt{u^2 + g^2 t^2} \rightarrow (8) \end{aligned}$$

This gives magnitude

And,

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{V_y}{V_x} \right) \\ &= \tan^{-1} \left(\frac{gt}{u} \right) \\ \therefore \theta &= \tan^{-1} \left(\frac{gt}{u} \right) \rightarrow (9) \end{aligned}$$

This gives direction.

Q. A body is projected horizontally from the top of tower 100 m ~~high~~ high with velocity of 9.8 m/s. Calculate

- ① Position of the body after 2 sec.
- ② Time taken by the body to reach the ground.
- ③ The distance from base of tower to the point where body strikes the ground.
- ④ The velocity with which body hits the ground.

→ Soln.

$$h = 100 \text{ m}$$

$$u = 9.8 \text{ m/s}$$

① $x = ?$, $y = ?$, When $t = 2 \text{ sec}$.

Now,

$$S_x = u_x t + \frac{1}{2} a_x t^2$$

$$\text{or, } x = u_x t + \frac{1}{2} \times 0 \times t^2$$

$$\text{or, } x = 9.8 \times 2$$

$$\text{or, } x = 19.6 \text{ m}$$

And,

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$\text{or, } y = 0 \times t + \frac{1}{2} g t^2 \quad \text{GSS_Notes}$$

$$\text{or, } y = \frac{1}{2} \times 9.8 \times 2^2$$

$$y = 19.6 \text{ m}$$

Hence, position of ~~proj~~ body is
19.6 m away and 19.6 m below
the top of tower.

② $T = ?$

We have formula,

$$T = \sqrt{\frac{2h}{g}}$$

$$T = \sqrt{\frac{2 \times 100}{9.8}}$$

$$T = 4.51 \text{ sec}$$

③ Horizontal Range (R) = ?

We know that,

$$R = uT$$

$$R = 9.8 \times 4.51$$

$$R = 44.27 \text{ m}$$

④ Final Velocity (V) = ?

We know that,

$$V = \sqrt{u^2 + g^2 t^2}$$

$$\text{or, } V = \sqrt{(9.8)^2 + (9.8 \times 4.51)^2}$$

$$\text{or, } V = 45.27 \text{ m/s}$$

$\therefore$ Final velocity = 45.27 m/s

This gives magnitude.

And,

$$\beta = \tan^{-1} \left(\frac{gt}{u} \right)$$

$$= \tan^{-1} \left(\frac{9.8 \times 4.51}{9.8} \right)$$

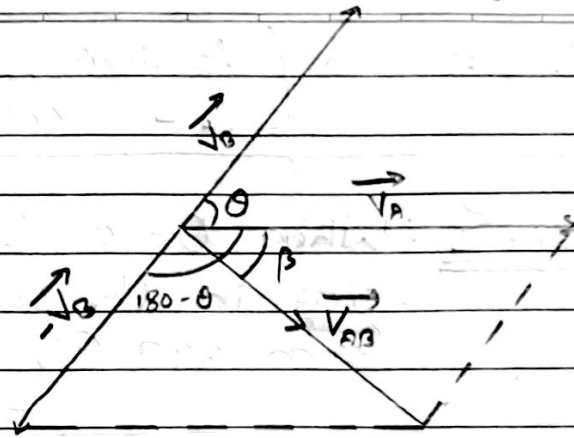
$$= \tan^{-1} (4.51)$$

$$\therefore \beta = 77.49^\circ$$

This gives direction.

* Relative Velocity

↳ The velocity of a body with respect to the moving body is called Relative velocity.



Consider two bodies A and B are moving with ~~velocity~~ velocities $\vec{V}_A$ and $\vec{V}_B$ respectively making an angle θ with each other.

Now, the relative velocity of A with respect to B is given by

$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$$

$$\therefore \vec{V}_{AB} = \vec{V}_A + (-\vec{V}_B)$$

Here, $\vec{V}_{AB}$ is the resultant of $\vec{V}_A$ and $-\vec{V}_B$ which is shown in fig.

Now,

according to Parallelogram law of vector addition the magnitude of V_{AB} is given by

$$V_{AB} = \sqrt{V_A^2 + (-V_B)^2 + 2V_A V_B \cos(180 - \theta)}$$

$$\therefore V_{AB} = \sqrt{V_A^2 + V_B^2 - 2V_A V_B \cos \theta} \rightarrow (i)$$

And,

direction of V_{AB} is given by

$$\beta = \tan^{-1} \left(\frac{V_B \sin(180 - \theta)}{V_A + V_B \cos(180 - \theta)} \right)$$

$$\therefore \beta = \tan^{-1} \left(\frac{V_B \sin \theta}{V_A - V_B \cos \theta} \right) \rightarrow (2)$$

Case-I: When $\theta = 0^\circ$ i.e. A and B are moving in the same direction then from eqn (1),

$$\begin{aligned} V_{AB} &= \sqrt{V_A^2 + V_B^2 - 2V_A V_B \cos 0} \\ \text{or, } V_{AB} &= \sqrt{V_A^2 + V_B^2 - 2V_A V_B \times 1} \\ \text{or, } V_{AB} &= \sqrt{V_A^2 - 2V_A V_B + V_B^2} \\ \text{or, } V_{AB} &= \sqrt{(V_A - V_B)^2} \\ \text{or, } V_{AB} &= V_A - V_B \quad \therefore V_{AB} = V_A - V_B \end{aligned}$$

Case-II: When $\theta = 180^\circ$ i.e. A and B are moving in the opposite directions then from eqn (1).

$$\begin{aligned} V_{AB} &= \sqrt{V_A^2 + V_B^2 - 2V_A V_B \cos 180^\circ} \\ \text{or, } V_{AB} &= \sqrt{V_A^2 + V_B^2 + 2V_A V_B} \\ \text{or, } V_{AB} &= \sqrt{(V_A + V_B)^2} \\ \text{or, } V_{AB} &= V_A + V_B \\ \therefore V_{AB} &= V_A + V_B \end{aligned}$$

Case-III: When $\theta = 90^\circ$ i.e. A and B are moving perpendicular to each other then From eqn (1).

$$\begin{aligned} V_{AB} &= \sqrt{V_A^2 + V_B^2 - 2V_A V_B \cos 90} \\ V_{AB} &= \sqrt{V_A^2 + V_B^2} \\ \therefore V_{AB} &= \sqrt{V_A^2 + V_B^2} \end{aligned}$$

And, from ②

$$\beta = \tan^{-1} \left[\frac{V_B \sin 90^\circ}{V_A - V_B \cos 90^\circ} \right]$$

$$= \tan^{-1} \left(\frac{V_B}{V_A} \right)$$

② Rain is falling vertically downward with velocity 4 km/hr and a boy is running on the horizontal road with velocity 3 km/hr.

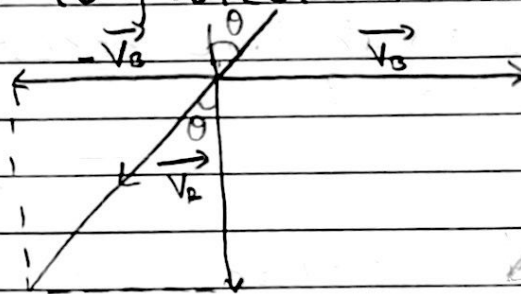
① Find relative velocity of rain with respect to the boy.

② At what angle should the boy tilt his umbrella in order to protect himself from the rain?

→ Soln,

$$V_A = 4 \text{ km/hr}$$

$$V_B = 3 \text{ km/hr}$$



① $V_{AB} = ?$

Here,

$$V_{AB} = \sqrt{(V_A)^2 + (V_B)^2}$$

$$= \sqrt{4^2 + 3^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25}$$

$$= 5 \text{ km/hr}$$

② $\theta = ?$

Here, $\tan \theta = \frac{V_A}{V_B}$

GSS_Notes

$$\text{or, } \tan \theta = \frac{3}{4}$$

$$\text{or, } \theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$\therefore \theta = 36.86^\circ$$

Q.10) A man can swim at the rate of 6 km/hr in still water. A river 1 km wide flows at the rate of 3 km/hr

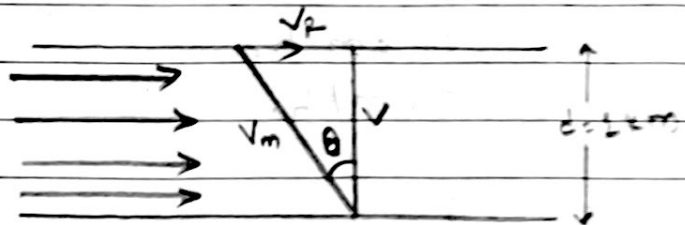
- ① Swimmer wishes to cross the river straight. ~~At~~ Along what direction must he strike?
- ② What will be his resultant velocity?
- ③ How much time he would take to cross the river?

→ Soln,

$$V_m = 6 \text{ km/hr}$$

$$V_r = 3 \text{ km/hr}$$

$$d = 1 \text{ km}$$



$$\text{① } \theta = ?$$

Here,

$$\sin \theta = \frac{V_r}{V_m}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

$$= \frac{1}{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\therefore \theta = 30^\circ$$

~~the must strike~~

Hence the swimmer should swim making an angle $30+90=120^\circ$ with the direction of flow of water.

② $V=?$

Here,

$$(V_m)^2 = (V_e)^2 + v^2$$

$$\text{or, } 6^2 = 3^2 + v^2$$

$$\text{or, } v^2 = 36 - 9$$

$$\text{or, } v = \sqrt{27}$$

$$\text{or, } v = 3\sqrt{3} \text{ km/hr}$$

$$\therefore v = 3\sqrt{3} \text{ km/hr}$$

③ Time Taken (T) = ?

Here,

$$V = \frac{d}{T}$$

$$\text{or, } 3\sqrt{3} = \frac{1}{T}$$

$$\text{or, } T = \frac{1 \text{ km}}{3\sqrt{3} \text{ km/hr}}$$

$$\text{or, } T = \frac{1}{3\sqrt{3}} \text{ hr}$$

$$\text{or, } T = \frac{1}{3\sqrt{3}} \times 60 \text{ min}$$

$$\therefore T = \frac{20}{\sqrt{3}} \text{ minute}$$

• Discussion Questions / Homework

Conceptual (short) Questions.

① Can an object with constant acceleration reverse its direction of travel? Explain.

→ Yes, An object moving upward under gravity has constant downward acceleration. Its velocity becomes zero at the highest point and then changes direction downward. Thus, even with constant acceleration, the direction of motion can reverse.

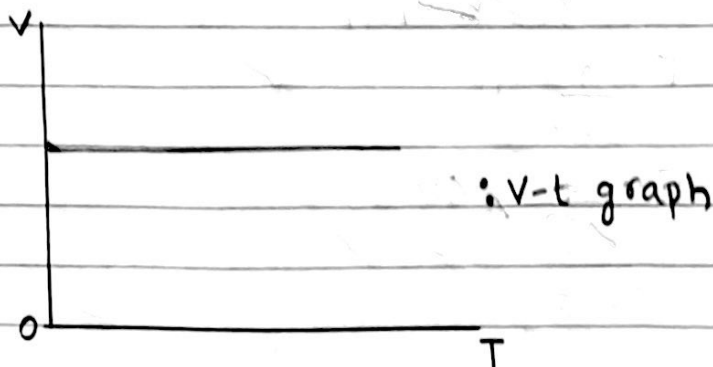
② Can a body have zero velocity but finite acceleration?

GSS_Notes

→ Yes, At the highest point of a vertically thrown ball, the velocity becomes zero, but acceleration due to gravity is still acting downward. Hence, velocity can be zero while acceleration is finite.

③ Can you have zero acceleration but non-zero velocity? Explain with a graph.

→ Yes, In uniform[✓] acceleration is zero but velocity is constant and non-zero.



④ Is it possible for a body to be accelerated even if it moves with uniform speed?

→ Yes, In circular motion, speed remains constant but direction changes continuously. Since velocity changes, the body has acceleration.

⑤ If the displacement of a body is proportional to square of time, show that body is moving with uniform acceleration.

→ Given,

$$s \propto t^2$$

$$s = kt^2$$

$$\frac{ds}{dt} = \frac{d(kt^2)}{dt}$$

$$\text{or, } v = 2kt$$

$$\text{or, } \frac{dv}{dt} = \frac{d(2kt)}{dt}$$

$$\text{or, } a = 2k$$

Since $a = 2k$ (constant) the body is moving with uniform acceleration.

⑥ A ball is dropped gently from the top of a tower and another ball is thrown horizontally at the same time. Which ball will hit the ground first?

→ Both ball will hit the ground at the same time. Their vertical motions are identical, and vertical motion is independent of horizontal motion.

⑦ From a flying aeroplane, a body should be dropped in advance to hit the target, why?

→ Because the body continues to move forward with the horizontal velocity of the aeroplane ~~but~~ while falling ~~into~~ vertically. To compensate for this horizontal motion, it must be released before reaching the target.

⑧ Find the angle of projection at which the horizontal range and maximum height of a projectile are equal.

→ Soln,

Horizontal range (R) = Max. Height (H_{max})

$$\frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

$$\text{or, } 2 \sin 2\theta = \sin^2 \theta$$

$$\text{or, } 4 \sin \theta \cdot \cos \theta = \sin^2 \theta$$

$$\text{or, } 4 \sin \theta \cdot \cos \theta - \sin^2 \theta = 0$$

$$\text{or, } \sin \theta (4 \cos \theta - \sin \theta) = 0$$

Either,

$$\sin \theta = 0$$

$$\theta = \sin^{-1}(0)$$

$$\theta = 0^\circ$$

$$4 \cos \theta = \sin \theta$$

$$\text{or, } \frac{\sin \theta}{\cos \theta} = 4$$

$$\text{or, } \tan \theta = 4$$

$$\text{or, } \theta = \tan^{-1}(4)$$

$$\therefore \theta = 0^\circ$$

⑨ A projectile moves in a parabolic path without air resistance. Is there any point at which its acceleration is perpendicular to the velocity? Explain.

→ Yes. At the highest point of the projectile, velocity is horizontal while acceleration due to gravity is vertical downward. Hence, acceleration is perpendicular to velocity at that point.

⑩ A projectile fired at an angle of 18° has certain horizontal range. State another angle of projection for the same horizontal range.

→ $\theta_1 = 18^\circ$

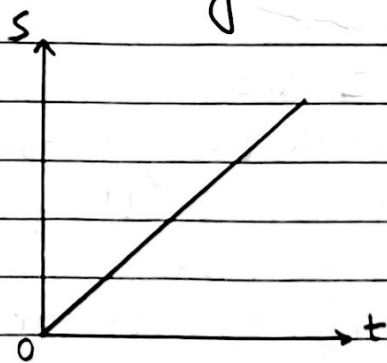
For same range

$$\theta_1 + \theta_2 = 90^\circ$$

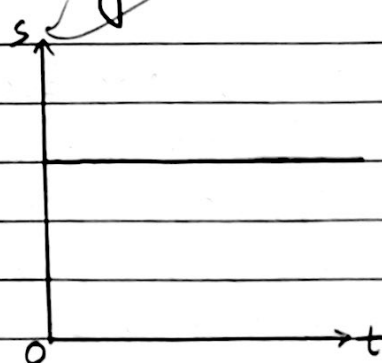
or, $\theta_2 = 90 - 18$

$$\therefore \theta_2 = 72^\circ \text{ } \frac{1}{2} \dots$$

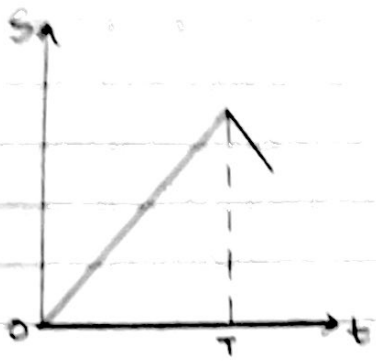
(ii) The slope of displacement time graph gives velocity of the body at any instant. State the nature of motion of the body in the following cases.



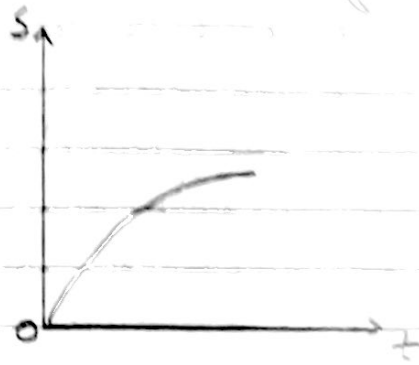
- Uniform positive Velocity



- Zero Velocity



- Non-uniform velocity
(accelerated and
decelerated motion)



- Non-uniform velocity
(decelerated motion)

- ⑫ A man walking in a rainy day must incline his umbrella at a certain angle though the rain is falling vertically down. Why?
- A man must incline his umbrella because, due to his motion, the rain appears to fall at an angle (relative velocity), even though it is falling vertically.

Numerical and Figure Based Questions

- ⑥ A body is projected horizontally from the top of a tower 100m high with a velocity of 9.8 m/s. calculate:
- Position of the body after 2 sec.
 - Time taken by body to reach the ground.
 - The distance from base of tower to the point where body strikes the ground.
 - The velocity with which body hits the ground.

→ Soln,

$$h = 100 \text{ m}$$

$$u = 9.8 \text{ m/s}$$

① $x = ?$, $y = ?$ When $t = 2 \text{ sec}$

$$S_x = u_x t + \frac{1}{2} a_x t^2$$

$$= 9.8 \times 2 + \frac{1}{2} \times \cancel{9.8} \times 0 + 2^2$$

$$= 19.6 \text{ m}$$

And,

$$S_y = \frac{1}{2} g t^2$$

$$= \frac{1}{2} \times 9.8 \times 2 \times 2$$

$$= 19.6 \text{ m}$$

Hence, position of body is 19.6m away and 19.6m below the top of the tower.

② $T = ?$

We have,

$$T = \sqrt{\frac{2h}{g}}$$

$$= \sqrt{\frac{2 \times 100}{9.8}}$$

$$= 4.51 \text{ sec}$$

③ Horizontal Range (R) = ?

We know that,

$$R = uT$$

$$= 9.8 \times 4.51$$

$$= 44.27 \text{ m}$$

④ $V = ?$

We know that,

$$V = \sqrt{u^2 + g^2 t^2}$$

$$= \sqrt{(9.8)^2 + (9.8 \times 4.51)^2}$$

$$= 45.27 \text{ m/s}$$

$$\therefore V = 45.27 \text{ m/s}$$

This gives magnitude,

And,

$$\theta = \tan^{-1} \left(\frac{gt}{u} \right)$$

$$= \tan^{-1} \left(\frac{9.8 \times 4.51}{9.8} \right)$$

$$= \tan^{-1} (4.51)$$

$$= 77.49^\circ$$

This gives direction.

⑤ A projectile is launched with an initial velocity of 30 m/s at an angle of 60° above the horizontal.

Calculate:

(i) position of projectile after 5s of its launch.

(ii) Magnitude and direction of its velocity 5s after the launch.

→ Soln,

$$u = 30 \text{ m/s}$$

$$\theta = 60^\circ$$

(i) $t = 5 \text{ s}$

$$x = ?, y = ?$$

We know that,

$$x = u \cos \theta \times t$$

$$\text{or, } x = 30 \times \cos 60 \times 5$$

$$= 30 \times \frac{1}{2} \times 5$$

$$= 15 \times 5$$

$$= 75 \text{ m}$$

$$\therefore x = 75 \text{ m}$$

And,

$$y = x \tan \theta - \left(\frac{g}{2 \cos^2 \theta} \right) x^2$$

$$= 75 (\tan 60) - \left(\frac{9.8}{32 \cos^2 60} \right) (75)^2$$

$$= 75 \sqrt{3} - 122.5$$

$$y = u \sin \theta \times t - \frac{1}{2} g t^2$$

$$= 30 \times \sin 60 \times 5 - \frac{1}{2} \times 9.8 \times 5^2$$

$$= 75 \sqrt{3} - 122.5$$

$$= 7.40 \text{ m}$$

Hence, position of body is 75m away and 7.40 m above the ground.

(ii) Magnitude of Velocity (V) is given by

$$\sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2}$$

$$\begin{aligned} &= \sqrt{(30 \cos 60)^2 + (30 \times \sin 60 - 9.8 \times 5)^2} \\ &= \sqrt{225 + (15\sqrt{3} - 49)^2} \\ &= \sqrt{225 + 529.88} \\ &= \sqrt{754.88} \\ &= 27.47 \text{ m/s} \end{aligned}$$

$\therefore$ Magnitude of Velocity (V) = 27.47 m/s.

And,

Direction of velocity is given by,

$$\beta = \tan^{-1} \left(\frac{u \sin \theta - gt}{u \cos \theta} \right)$$

$$= \tan^{-1} \left(\frac{30 \times \sin 60 - 9.8 \times 5}{30 \times \cos 60} \right)$$

$$= \tan^{-1} \left(\frac{-23.01}{15} \right)$$

$$= -56.90$$